ANALYTICS WHITEPAPER

CLUTCH-X

A reproducible methodology for T20 pressure analytics

A transparent, reproducible framework for identifying T20 players who execute above expectation in high-leverage moments.

Dataset snapshot: 2022 to 2026. 193875 deliveries across 831 matches and 136 competitions.

DELIVERIES	MATCHES	ELIGIBLE BATTERS	ELIGIBLE BOWLERS
193875	831	47	68

RANKING STABILITY: BELOW TARGET
The PRD requires split-half Spearman rho >= 0.65. The measured result is batting rho 0.1985 and bowling rho 0.3317. Rankings are exploratory and must not be presented as validated stable rankings.

01 / DATA AND PRESSURE AUDIT

Coverage and pressure separation

The analysis scopes competition data to IPL and T20I, reconstructs the pre-delivery match state, and excludes Super Overs and abandoned/no-result matches with insufficient play. The dataset contains 16.9% high-pressure deliveries at the PI >= 60 cutoff.

PRESSURE CHECK	OBSERVED	ACCEPTANCE
High-pressure share	16.9%	15% to 25%
High-pressure wicket rate	8%	Compare regimes
Normal-pressure wicket rate	4.9%	Reference
Wicket-rate chi-square p	0.0000000000


25-29 1792
30-34 6432
35-39 1153
40-44 889
45-49 5671
50-54 778
55-59 525
60-64 5330
65-69 563
70-74 3532
75-79 10199
80-84 1538
85-89 2454
90-94 7077
95-99 2037

02 / SCORING METHODOLOGY

How pressure and performance are measured
Delivery-level Pressure Index

Second innings: chase tension is 100 x clip(RRR / par rate - 1, 0, 1)^2, weighted 75%; resource scarcity ((wickets lost / 9) x (1 - balls remaining / scheduled balls)) is weighted 25%. **First innings:** PI is the greater of the early-collapse trigger and death-over urgency. Knockouts receive a 1.15 multiplier, then PI is clamped to 0-100. High pressure starts at PI 60; critical leverage starts at PI 80. Collapse pressure starts at 70 for three wickets lost by over 8 (+5 per additional wicket), or 75 for five wickets by over 14 (+4 per wicket). Death-over urgency starts at 30 in over 16 and increases 15 per over.

Context baselines and reliability Expected delivery outcomes are bucketed by format, season, innings, and match phase; sparse buckets shrink toward the parent expectation using 30 prior deliveries. A player needs at least 150 high-pressure balls and 10 high-pressure matches. Per-player pressure metrics shrink toward the eligible peer mean with reliability weight

w = pressure balls / (pressure balls + 150)

Match-to-match consistency is adjusted as mean impact minus 0.5 times the sample standard deviation. Batting impact is scaled per 100 balls; bowling runs saved is scaled per six legal balls.

Rank construction Each component is percentile-ranked within its eligible discipline cohort. Baseline ratings weight ordinary-condition performance; pressure ratings weight reliability-adjusted pressure performance. The final score is:

Clutch-X = 0.55 x pressure rating
1. 0.25 x clip((net clutch delta + 100) / 2, 0, 100)
2. 0.20 x positive-impact rate (as a percentage).

Batting baseline: 60% normal strike-rate percentile, 20% boundary-rate percentile, 20% dot-ball-avoidance percentile. **Pressure rating:** 40% volatility-adjusted pRAE, 25% shrunk strike-rate delta, 10% boundary-rate delta, 10% dot-rate improvement, 10% wicket preservation, 5% positive-impact rate.

Bowling baseline: 60% economy percentile, 20% dot-rate percentile, 20% wickets-per-24-balls percentile. **Pressure rating:** 40% volatility-adjusted pRSAE, 25% shrunk economy delta, 10% dot-rate improvement, 20% clutch-wicket impact, 5% positive-impact rate.

03 / STABILITY AUDIT

Acceptance evidence

The odd/even chronological match split is evaluated on eligible players only; each half is volume-weighted before Spearman correlation. This dataset misses the PRD acceptance threshold in both disciplines.

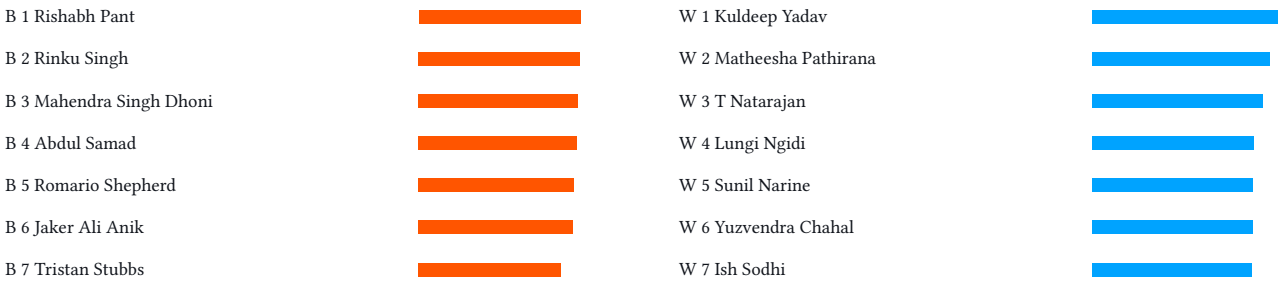
DISCIPLINE	ELIGIBLE	SPLIT rho	TARGET	STATUS
Batting	47	0.1985	0.65	BELOW TARGET
Bowling	68	0.3317	0.65	BELOW TARGET

04 / VERIFIED RANKINGS

Batting Top 10

These rows are read directly from the deterministic `top10_batters.parquet` artifact. Ranks and player IDs are retained without recalculation or re-sorting in the report layer.

Clutch-X score comparison (0-100)



B 8 Dasun Shanaka

B 9 Towhid Hridoy

B 10 Suryakumar Yadav

W 8 Shaheen Afridi

W 9 Mohammad Rishad Hossain

W 10 Harshit Rana

RANK	PLAYER / ID	HP / BALLS MATCHES	SR BASE -> HP	SR DELTA	pRAE / 100	POSITIVE	BASE -> PRESSURE	SCORE
1	Rishabh Pant (4703)	166 / 23	135.3 -> 194	+58.7	+42.4	69.6%	46.1 -> 81.7	**75.8**
2	Rinku Singh (4708)	472 / 52	128.6 -> 193.6	+65.1	+45.3	57.7%	37.8 -> 83.4	**75.6**
3	Mahendra Singh Dhoni (8)	279 / 41	88 -> 190.3	+102.3	+37.5	56.1%	1.7 -> 75.6	**74.6**
4	Abdul Samad (7446)	199 / 26	129.3 -> 172.9	+43.5	+21.9	57.7%	33.5 -> 80.7	**74.3**
5	Romario Shepherd (6726)	396 / 52	128.5 -> 173	+44.5	+32.3	57.7%	27.4 -> 77.1	**72.7**
6	Jaker Ali Anik (4773)	169 / 17	120.4 -> 149.7	+29.3	+14.2	58.8%	13 -> 73.7	**72.4**
7	Tristan Stubbs (20702)	320 / 34	135.5 -> 182.8	+47.3	+39.5	47.1%	51.3 -> 75.8	**66.7**
8	Dasun Shanaka (2436)	237 / 34	120.1 -> 183.5	+63.4	+41.7	47.1%	10.9 -> 67.8	**66.3**
9	Towhid Hridoy (6651)	158 / 12	120.3 -> 153.2	+32.9	+17.7	50%	19.6 -> 68.5	**66.3**
10	Suryakumar Yadav (1699)	226 / 28	163.4 -> 198.2	+34.8	+52.2	57.1%	87 -> 75.6	**64.1**

04 / VERIFIED RANKINGS

Bowling Top 10

These rows are read directly from the deterministic `top10_bowlers.parquet` artifact.

RANK	PLAYER / ID	HP BALLS / MATCHES	ECONOMY BASE -> HP	ECONOMY DELTA	pRSAE / OVER	WICKETS / POSITIVE	BASE -> PRESSURE	SCORE
1	Kuldeep Yadav (3644)	221 / 33	8.28 -> 6.03	+2.2	+2.7	23 / 87.9%	59.7 -> 95.1	**86.8**
2	Matheesha Pathirana (8661)	365 / 34	9.08 -> 8.37	+0.7	+2.2	40 / 67.6%	37 -> 91.4	**83.1**
3	T Natarajan (4265)	340 / 36	9.94 -> 9	+0.9	+1	29 / 55.6%	4.8 -> 84.6	**80.1**
4	Lungi Ngidi (4641)	170 / 26	8.97 -> 8.86	+0.1	+1.1	19 / 57.7%	37.9 -> 83.3	**75.5**
5	Sunil Narine (781)	186 / 18	7 -> 5.94	+1.1	+3.1	11 / 88.9%	69 -> 79.4	**75.2**
6	Yuzvendra Chahal (1612)	240 / 36	8.69 -> 8.05	+0.6	+1.4	21 / 69.4%	42.4 -> 80.1	**75.2**
7	Ish Sodhi (2526)	150 / 18	8.73 -> 7.44	+1.3	+0.8	11 / 66.7%	37 -> 79.2	**74.6**
8	Shaheen Afridi (6355)	180 / 28	8.17 -> 7.4	+0.8	+2.2	18 / 71.4%	77.9 -> 78.9	**70.3**
9	Mohammad Rishad Hossain (8380)	175 / 18	8.7 -> 7.47	+1.2	+0.8	11 / 61.1%	50.4 -> 76	**69.7**
10	Harshit Rana (29888)	168 / 23	9.6 -> 9.43	+0.2	+0.5	13 / 69.6%	38.8 -> 70.9	**69.4**

05 / EXPOSURE AND PROVENANCE

Audit trail

Top-10 high-pressure exposure is summarized by competition and innings; it is descriptive and does not impose a quota on the rank order.

DISCIPLINE	COMPETITION	INNINGS	HP VOLUME / PLAYERS
Batting	IPL	1	977 / 8
Batting	IPL	2	380 / 8
Batting	T20I	1	960 / 8
Batting	T20I	2	305 / 6
Bowling	IPL	1	1000 / 7
Bowling	IPL	2	322 / 7
Bowling	T20I	1	655 / 8
Bowling	T20I	2	218 / 8

Ranking artifact SHA-256 top10_batters.parquet 51be20d124eb773be67c65bd0d75b22acf2ebe04758147e37da40140f1e22b1d
top10_bowlers.parquet 9749c09afd240f407a171a6202f87d63c8c6b611195b53a41e78d17e6340298c
ranking_audit_summary.parquet eaee7dc98f7cc77795c15928f70c82adb70275d28163b75a0adf85910d6e1205

Source CSV SHA-256 ball_by_ball.csv 01e2b4379edfbab6a6e456d304cb3c70f554cb312ef93d9588f3f32a82cb396e
matches.csv bb100ab84b8b9a15160c1f2315cf675210131a30e779ec5079fd5403dcf56eef
competitions.csv 507ceb7189f972d4e4f299ddcfc97f3c3247af1913f6291106f161fa8353fd9b
playing_eleven.csv 7517c73722a28d5d61c808ccf0cbed315a8780586d02e946eb9a7f5daf011c27
teams.csv a65e5922a9b3bf49247b24fc9471782e700efb31f8dcc5042923238b87c50a5d
competition_team_player_mapping.csv 1ded4e42a933618572d6e2c3a521c3257eb1844fab4456d567d77cbb2d3bbb12

06 / LIMITATIONS

Interpretation and responsible use

The available data does not encode dew, boundary dimensions, field placements, pitch deterioration, weather, or all tactical choices. Competition coverage and player exposure are uneven; the pressure index and contextual baselines are deterministic proxies, not causal estimates. Match context and role changes can influence observed performance. Shrinkage and volatility penalties reduce but do not eliminate sampling bias. The split-half stability gate is not met (rho target >= 0.65); use these rankings as exploratory evidence, not as a validated measure of future performance. The threshold is reported unchanged.

Reproduction: `python -m clutch_x.report.build_report`. The report data JSON, Typst source, and PDF are emitted together. The report builder validates rank integrity before compiling and hashes the exact ranking artifacts used.